

John (“Jack”) Klawitter

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EDUCATION

Rutgers University

Doctor of Philosophy (Ph.D.) in Computer Science (Enrolled)

- Advisor: Dr. Konstantinos Michmizos | Computational Brain Lab, CBIM

New Brunswick, NJ

Sep. 2023 – Present

Middlebury College

Bachelor of Arts with Majors in Physics and Computer Science, Magna Cum Laude

Middlebury, VT

Sep. 2018 – Feb. 2023

EXPERIENCE

Graduate Research Assistant

Rutgers University

Sep. 2023 – Present

New Brunswick, NJ

- Developed interpretable machine learning models for decoding high-dimensional neural recordings, enabling analysis of representational drift across motor learning
- Introduced Critical Spike Attribution, a novel feature attribution method for spiking neural networks, providing causal insight into spike-level contributions to model predictions
- Applied representation analysis and feature attribution to align model dynamics with biological change
- Led collaboration between Computer Science and Psychology labs to integrate modeling and experimental data collection

Physics Research Assistant

Middlebury College

Feb. 2022 – Dec. 2022

Middlebury, VT

- Developed Python script to correct the spectra of red quasars
- Published open-source code for use by the astrophysics community

Machine Learning Intern

Rolls-Royce

May 2022 – August 2022

Indianapolis, IN

- Developed deep learning models for early detection of jet engine failures, improving detection rate by approximately 10% in operational data

Undergraduate Researcher

Ithaca College Dynamical Systems REU

May 2021 – July 2021

Ithaca, NY

- Studied billiard dynamics on surfaces of revolution
- Developed a Mathematica notebook that models billiard dynamics on any surface of revolution within a “billiard board” — any region bounded by a piecewise-defined curve
- Developed original billiard constructions, including a new “unilluminable room” on the sphere and a novel result regarding the existence of periodic orbits on surfaces of revolution.
- Presented findings at the 2022 Joint Mathematics Meeting poster session

Math Research Assistant

Middlebury College

May 2020 – August 2020

Middlebury, VT

- Worked on Martin Gardner’s “Minimum No-3-In-A-Line” Problem
- Examined the use of SAT Solvers as they apply to combinatorial problems
- Modified programs written by Dr. Donald Knuth and created novel Python scripts

PUBLICATIONS

- Klawitter, J., Michmizos, K. **Critical Spike Attribution for Feature Importance in Spiking Neural Networks.** *Neuro-Inspired Computational Elements (NICE)*, 2026 (Oral Presentation)
- Klawitter, J., Michmizos, K. **Sequential Learning in Neural Decoders Reveals Interpretable Neural Representations.** *Neurocomputing*, 2026. 132742, ISSN 0925-2312.

AWARDS & HONORS

Rutgers Brain Health Institute Trainee Travel Award, Spring 2026

TEACHING

Course Instructor

Design and Analysis of Computer Algorithms — Summer 2024

Teaching Assistant

Intro to Discrete Structures I — Spring 2024, Spring 2025

Intro to Discrete Structures II — Fall 2024

Intro to Artificial Intelligence — Summer 2025

SKILLS

- **Programming:** Python (NumPy, PyTorch, scikit-learn)
- **Machine Learning:** Deep learning, spiking neural networks, neural decoding, time-series modeling, representation analysis, feature attribution, interpretable ML
- **Tools:** Git, LaTeX